Work and Energy

The big idea

In science, **work** is done only when a force **moves** an object. Energy is the **capacity to do work**, and it is never lost, only changed from one form to another.

Work

WORK DONE BY A CONSTANT FORCE

W = F × s (in the direction of the force); unit joule (J): 1 J = 1 N × 1 m

Work is **positive** when force and displacement are in the same direction (pushing a cart forward), **negative** when opposite (friction slowing a ball), and **zero** when they are perpendicular or there is no displacement (a porter carrying a load horizontally does no work against gravity; pushing a wall does no work).

Kinetic and potential energy

TWO FORMS OF MECHANICAL ENERGY

Kinetic energy (energy of motion): KE = ½mv²

Potential energy (energy of position, height h): PE = mgh

A 2 KG BALL MOVES AT 5 M/S. FIND ITS KINETIC ENERGY.

KE = ½ × 2 × 25 = **25 J**.

Doubling the speed makes KE **four times** larger. Potential energy depends on height, not the path taken to get there.

Conservation of energy

LAW OF CONSERVATION OF ENERGY

Energy can neither be created nor destroyed; it can only be transformed from one form to another. The **total energy stays constant**.

A FREELY FALLING BALL

At the top, all its energy is PE. As it falls, PE turns into KE. Just before hitting the ground, it's all KE. At every point, **PE + KE = constant** (ignoring air resistance).

Other transformations: chemical energy in food → muscles; electrical → light and heat in a bulb; solar → chemical in photosynthesis.

Power

POWER

P = W / t (rate of doing work); unit watt (W): 1 W = 1 J/s
Commercial unit of energy: 1 kWh = 3.6 × 10^6 J (one "unit" on the electricity bill)

A 60 W BULB RUNS FOR 10 HOURS. ENERGY USED?

60 × 10 = 600 Wh = **0.6 kWh**.

Quick recap

- W = Fs; it can be positive, negative or zero.
- KE = ½mv², PE = mgh.
- Total energy is conserved.
- Power = work/time; 1 kWh = 3.6 MJ.

Practice questions

1. A coolie carries a load on his head and walks on a level road. The work done against gravity is
   (a) zero (b) positive (c) negative (d) infinite
2. If the velocity of a body is doubled, its kinetic energy becomes
   (a) double (b) half (c) four times (d) the same
3. The potential energy of a 10 kg object at a height of 5 m is ($g = 10\ m/s^2$)
   (a) 50 J (b) 500 J (c) 5000 J (d) 15 J
4. 1 kWh equals
   (a) 1000 J (b) 3600 J (c) 3.6×10^6 J (d) 3.6×10^3 J
5. A machine does 1200 J of work in 1 minute. Its power is
   (a) 1200 W (b) 20 W (c) 72000 W (d) 2 W

Answers: 1. a 2. c 3. b 4. c 5. b